

SECOND SEMESTER B.A./B.Sc. DEGREE EXAMINATION, APRIL 2020

(CBCSS—UG)

Mathematics

MTS 2B 02—CALCULUS OF SINGLE VARIABLE—I

(2019 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes**Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 2B 02—CALCULUS OF SINGLE VARIABLE—I

(Multiple Choice Questions for SDE Candidates)

1. Given $f(x) = 3x$ and $g(x) = x^2 - 1$. Then, the domain of $\frac{f}{g}$ is:
- (A) $[1, \infty)$. (B) $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.
(C) $(0, \infty)$. (D) $(1, \infty)$.
2. $\lim_{x \rightarrow \infty} \sin(x)$ is:
- (A) 0. (B) 1.
(C) 1 or -1. (D) Limit does not exist.
3. If f is continuous at every point of a closed interval I , then f assumes:
- (A) An absolute maximum value M but not an absolute minimum value.
(B) An absolute minimum value m but not an absolute maximum value.
(C) Both an absolute maximum value M and an absolute minimum value m .
(D) Neither an absolute maximum nor an absolute minimum.
4. On $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $f(x) = \sin(x)$ takes on:
- (A) A maximum value of 1 (once) and a minimum value of 0 (twice).
(B) A maximum value of 1 (once) and minimum value of -1.
(C) A maximum value of 1 (once) and no minimum value.
(D) A minimum value of -1 (once) and no maximum value.
5. The only domain points where a function can assume extreme values are _____.
- (A) Critical points and end points. (B) Critical points only.
(C) End points only. (D) None of the above.
6. Using which of the following reasons, can we conclude that "The Rolle's theorem cannot be applied to the function $f(x) = \tan x$ for the interval $[0, \pi]$."
- (i) There is a discontinuity at $x = \frac{\pi}{2}$ to the function $f(x) = \tan x$.
(ii) $f'(x) = \sec^2 x$ which does not exist at $x = \frac{\pi}{2}$.
- (A) Both (i) and (ii). (B) (i) only.
(C) (ii) only. (D) None of the above.

7. The value or values of c that satisfy the equation $\frac{f(b) - f(a)}{b - a} = f'(c)$ in the conclusion of Mean Value Theorem for the function $f(x) = x^2 + 2x - 1$ and the interval $[0, 1]$ is:
- (A) 1. (B) $\frac{1}{2}$.
 (C) $\frac{1}{3}$. (D) $\frac{1}{4}$.
8. At a critical point c , if f' changes from positive to negative at c ($f' > 0$ for $x < c$ and $f' < 0$ for $x > c$), then f has _____.
- (A) A local maximum value at c . (B) A local minimum value at c .
 (C) Global minimum value at c . (D) None of the above.
9. A curve is said to be concave upwards (or convex downwards) at or near P when at all points near P on it _____.
- (A) Lies *above* the tangent at P . (B) Lies *below* the tangent at P .
 (C) Lies *on* the tangent at P . (D) None of these.
10. $\lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right) =$ _____.
- (A) 0. (B) $\frac{1}{5}$.
 (C) 5. (D) ∞ .
11. $\lim_{x \rightarrow -\infty} \frac{11x + 2}{2x^3 - 1} =$ _____.
- (A) 0. (B) $\frac{11}{2}$.
 (C) ∞ . (D) $-\infty$.
12. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(4x)} :$
- (A) 1. (B) $\frac{4}{3}$.
 (C) $\frac{3}{4}$. (D) 0.
13. $\lim_{x \rightarrow 0} \frac{\sin(x)}{x^2 + 3x} :$
- (A) 1. (B) $\frac{1}{3}$.
 (C) 3. (D) $\frac{1}{4}$.

Turn over

14. Use the linear approximation of $f(x) = \sqrt{1+x}$ at $a = 0$ to estimate $\sqrt{0.95}$:
- (A) 0.942. (B) 0.995.
(C) 0.9820. (D) 0.9750.
15. If $x^2 + 2xy = y^2$, then $\frac{dy}{dx}$ is :
- (A) $\frac{x+y}{y-x}$. (B) $2x + 2y$.
(C) $\frac{x+1}{y}$. (D) $-x$.
16. If $y = 9x^2 - 4x + 3$, then $\frac{d^2y}{dx^2}$ is :
- (A) $18x - 4$. (B) -4 .
(C) 22 . (D) 18 .
17. Determine the extremas of the following function $4x^3 - 48x$:
- (A) $(2, -64)$. (B) $(2, -64)$ and $(-2, 64)$.
(C) $(2.3, -61.9)$ and $(-2, 64)$. (D) $(-2.3, 61.9)$ and $(2, -64)$.
18. The linearization of $f(x) = x^3$ at $x = 2$ is _____.
- (A) $2(6x - 7)$. (B) $2(6x + 7)$.
(C) $3x$. (D) 0 .
19. $d(\cot u) =$ _____.
- (A) $-\operatorname{cosec}^2 u \, du$. (B) $\operatorname{cosec}^2 u \, du$.
(C) $\sin u \, du$. (D) $-\sec^2 u \, du$.
20. The radius r of a circle increases from $r_0 = 10 \, m$ to $10.1 \, m$. Estimate the increase in the circle's area A by calculating dA :
- (A) $dA = 2\pi m^2$. (B) $dA = -2\pi m^2$.
(C) $dA = \pi m^2$. (D) $dA = -\pi m^2$.

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Mathematics

MTS 2B 02—CALCULUS OF SINGLE VARIABLE—I

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer any number of questions.**Each question carries 2 marks.**Maximum 25 marks.*

1. Let f and g be functions defined by $f(x) = x + 1$ and $g(x) = \sqrt{x}$. Find the functions $g \circ f$ and $f \circ g$. What is the domain of $g \circ f$?
2. Find $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$.
3. Let $f(x) = \begin{cases} x^2 - x - 2 & \text{if } x \neq 2 \\ 1 & \text{if } x = 2. \end{cases}$ Show that f has a removable discontinuity at 2. Redefine f at 2 so that it is continuous everywhere.
4. Find $\lim_{x \rightarrow \pi/4} \frac{\sin x}{x}$.
5. Show that $f(x) = |x|$ is continuous everywhere.
6. Find the derivative of $\sqrt[3]{x} + \frac{1}{\sqrt{x}}$.
7. Find the critical points of $f(x) = x^3 - 6x + 2$.
8. Find $\lim_{x \rightarrow \infty} (2x^3 - x^2 + 1)$ and $\lim_{x \rightarrow -\infty} 2x^3 - x^2 + 1$.
9. Find the interval on which $f(x) = x^2 - 2x$ is increasing or decreasing.
10. Find the vertical asymptote of the graph of $f(x) = \frac{1}{x-1}$.

Turn over

11. Find $\int \frac{\cos x}{1 - \cos^2 x} dx$.
12. Find $\int x e^{-x^2} dx$.
13. Suppose $\int_1^6 f(x) dx = 8$ and $\int_4^6 f(x) dx = 5$, what is $\int_1^4 f(x) dx$.
14. Find the volume of the solid obtained by revolving the region under the graph of $y = \sqrt{x}$ on $[0, 2]$ about the x -axis.
15. Find the work done in lifting a 2.4 kg. package 0.8 m. off the ground (given $g = 9.8 \text{ m./sec.}^2$).

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 35 marks.

16. Find the slope and an equation of the tangent line to the graph of the equation $y = -x^2 + 4x$ at the point $P(2, 4)$.
17. Suppose that $g(x) = (x^2 + 1)f(x)$ and it is known that $f(2) = 3$ and $f'(2) = -1$. Evaluate $g'(2)$.
18. (a) Show that $f(x) = x^3$ satisfies the hypothesis of the mean value theorem on $[-1, 1]$.
 (b) Find the numbers c in $(-1, 1)$ that satisfies the equation as guaranteed by the mean value theorem.
19. Find the slant asymptotes of the graph of $f(x) = \frac{2x^2 - 3}{x - 2}$.
20. A car moves along a straight road with velocity function $v(t) = t^2 + t - 6$, $0 \leq t \leq 10$, where $v(t)$ is measured in feet per second.
 (a) Find the displacement of the car between $t = 1$ and $t = 4$.
 (b) Find the distance covered by the car during this period.
21. (a) Evaluate $\int_{-3}^0 (x^2 - 4x + 7) dx$ by Fundamental theorem of Calculus.
 (b) Use the definition of definite integral to show that if $f(x) = c$, a constant function, then $\int_a^b f(x) dx = c(b - a)$.
22. Find the center of mass of a system comprising three particles with masses 2, 3 and 5 slugs, located at the points $(-2, 2)$, $(4, 6)$ and $(2, -3)$ respectively.
23. Find the length of the graph of $x = \frac{1}{3}y^3 + \frac{1}{4y}$ from $P\left(\frac{7}{12}, 1\right)$ to $G\left(\frac{67}{24}, 2\right)$.

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. (a) Find $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta}$.
- (b) Use intermediate value theorem to find the value of c such that $f(c) = 7$, where $f(x) = x^2 - x + 1$ on $[-1, 4]$.
- (c) In a fire works display, a shell is launched vertically upwards from the ground, reaching a height $S = -16t^2 + 256t$ feet after t seconds. The shell burst when it reaches its maximum height :
- (i) A what time after launch will the shell burst.
- (ii) What will be the altitude of the shell when it explodes ?
25. Find the dimensions of the rectangle of greatest area that has its base on the x -axis and is inscribed in the parabola $y = 9 - x^2$.
26. Using the definition of definite integral evaluate $\int_a^b x dx$.
27. Find the area of the surface obtained by revolving the graph of $f(x) = \sqrt{x}$ on the interval $[0, 2]$ about the x -axis.

$(2 \times 10 = 20 \text{ marks})$